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Measuring Neural-Network Robustness Under Adversarial Perturbation

Experimental Evaluation of Model Behavior Under Gradient-Based Attacks

Source inventory: f7405a395ea35ded09825ff2b88ee3a8e40a834c

Test snapshot: 490b71ec480c40622b63b1adaf2bf2f75959b8d6

CI run 37169514061 | initiated 2026-10-04 UTC | MIT

https://github.com/poojakira/adversarial-ml-lab

112
tests passed in cited CI

32.22%
statement coverage at cited CI

71.82%
reported clean CIFAR-10 accuracy

0.00%
reported PGD-20 robust accuracy, epsilon 8/255

CONTRIBUTION

- A reproducible robustness-evaluation harness for FGSM, PGD, and C&W that shows a measured 71.82% clean CIFAR-10 SmallCNN collapsing to 0.00% robust accuracy under PGD-20 at epsilon 8/255, while explicitly separating measured results from literature projections.

IMPLEMENTED METHOD

- 1. Train/evaluate a compact CIFAR-10 classifier under a fixed seed and CPU budget.
- 2. Measure clean accuracy on the full CIFAR-10 test set.
- 3. Generate epsilon-bounded FGSM and PGD adversarial samples plus C&W L2 samples.
- 4. Measure robust accuracy on the committed attack subset.
- 5. Emit structured evidence and CI robustness-gate logic.

LIMITATIONS AND EVIDENCE BOUNDARY

- This repository measures vulnerability; it does not make a model robust.
- Robust-accuracy attacks use a 1,024-sample subset, not the full 10,000 test set.
- No AutoAttack, black-box transfer, or certified-robustness result is established.
- results/cifar10_resnet18_benchmark.json is explicitly literature-projected/synthetic and must not be presented as measured.
- Torch/dataset availability blocks a fresh model reproduction; CI test passing is separate from the historical robustness experiment.

SCOPED EVIDENCE

- Cited Python 3.12 CI: 112 passed; 32.22% statement coverage, same test job.
- Separate committed measurement 1282d78c42cfcd6a6ecd70b711d95058cb3ad7b0, September 3, 2026: SmallCNN has 1,117,354 parameters, 6 CPU epochs, seed 42; clean evaluation uses 10,000 images.
- Actual attack subset: 1,024 samples (requested 1,000). FGSM robust accuracy 3.32%; PGD-20 0.00% at epsilon 8/255 and step 2/255; C&W-L2 4.20% (c=1, 100 steps).
- Source: results/cifar10_smallcnn_real.json. The cited October CI robustness job was skipped; these historical model measurements were not rerun locally.

REPRODUCTION

- git checkout 490b71ec480c40622b63b1adaf2bf2f75959b8d6
- Install the documented development dependencies; run python -m pytest tests/ -q. Use the cited job configuration when reproducing coverage.
- Benchmark data, adapters and environment requirements are separate; consult poster/13_reproducibility.md and the claim ledger.